

FPGA-Based Adaptive Voltage Window Protection System

Team Members:

Visham Kumar V – CB.EN.U4EEE23047

Pranav Krishna JS – CB.EN.U4EEE23023

Devendra Parandaman – CB.EN.U4EEE23147

Shaaron VG – CB.EN.U4EEE23033

1. Introduction

Voltage monitoring and protection systems are essential in power electronics and embedded applications to ensure safe operation of electronic systems. In digital protection systems, threshold-based window detection is widely used to identify abnormal voltage conditions such as over-voltage and under-voltage.

This project implements a **Voltage Window Protection System** using an FPGA (Artix-7 XC7A100T). The system detects whether an input voltage sample lies within a predefined window around a nominal reference value. A Finite State Machine (FSM) is used to classify the system into NORMAL, OVER, or UNDER states. Additionally, a Freeze-on-Fault mechanism is incorporated to ensure stable and deterministic system recovery.

The design is implemented using VHDL and validated through simulation, RTL synthesis, floorplanning, and power analysis in Vivado.

2. Objectives

- To design a voltage window detection system using digital logic.
- To implement over-voltage and under-voltage detection using comparators.
- To design a synchronous FSM-based control system.
- To implement Freeze-on-Fault logic for stable operation.
- To verify the design using simulation and hardware implementation.
- To analyze FPGA resource utilization and power consumption.

3. Theoretical Background

3.1 Voltage Window Detection

In a voltage window detection system:

Upper Threshold:

$$V_HIGH = V_REF + \Delta$$

Lower Threshold:

$$V_LOW = V_REF - \Delta$$

Where:

- V_REF = Nominal reference voltage (230)
- Δ = Tolerance margin (20)

System Conditions:

- If $V_LOW \leq V \leq V_HIGH \rightarrow$ NORMAL
- If $V > V_HIGH \rightarrow$ OVER-VOLTAGE
- If $V < V_LOW \rightarrow$ UNDER-VOLTAGE

This concept is derived from comparator-based digital protection circuits studied in Digital Electronics.

3.2 Finite State Machine (FSM)

The system is implemented as a **synchronous Moore FSM** with three states:

1. NORMAL
2. OVER
3. UNDER

State transitions occur on the rising edge of the clock. The next state depends on comparator outputs.

FSM ensures:

- Deterministic behavior
 - Stable transitions
 - Clean synchronous design
 - No latches inferred
-

3.3 Freeze-on-Fault Logic

A key enhancement in this design is the Freeze-on-Fault mechanism.

In conventional window systems, the reference value may drift during abnormal conditions. In this design:

- When a fault (OVER or UNDER) occurs,
- Reference update mechanism is disabled,
- Window thresholds remain fixed,
- System recovers only when voltage returns to safe range.

This prevents oscillatory behavior and threshold shifting.

4. System Architecture

Block Diagram

Switch Input → Comparator → FSM → Freeze Logic → LED Output

Functional Blocks

1. Input Block
 - 8-bit switch input represents voltage sample.
 2. Comparator Block
 - Compares input with V_HIGH and V_LOW.
 3. FSM Control Block
 - Determines system state.
 4. Freeze Logic
 - Prevents threshold drift during faults.
 5. Output Block
 - LED[2] → OVER (Red)
 - LED[1] → NORMAL (Green)
 - LED[0] → UNDER (Blue)
-

5. Design Implementation

5.1 VHDL Design

The design is implemented using:

- Synchronous process (rising_edge(clk))
- Case statement for FSM
- Integer arithmetic for threshold calculation

- Combinational LED mapping

The system uses:

- One clock input (100 MHz)
- One reset input
- 8-bit voltage input
- 3-bit LED output

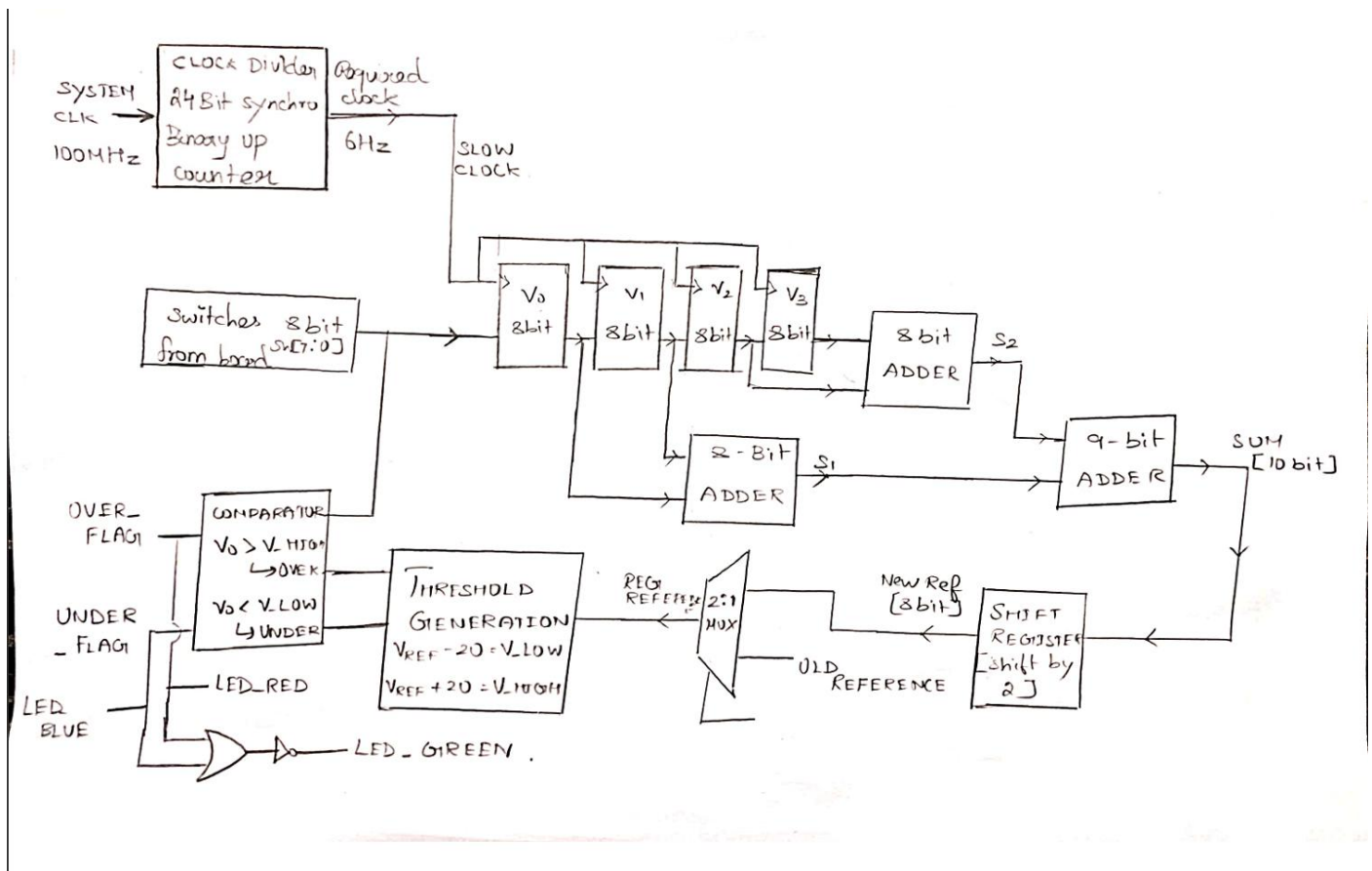
No asynchronous feedback loops are present.

5.2 State Encoding

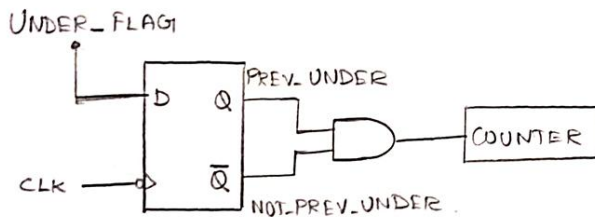
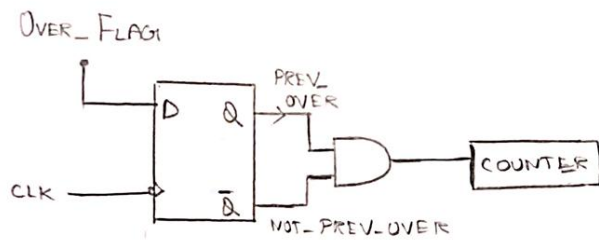
States are defined as:

- NORMAL
- OVER
- UNDER

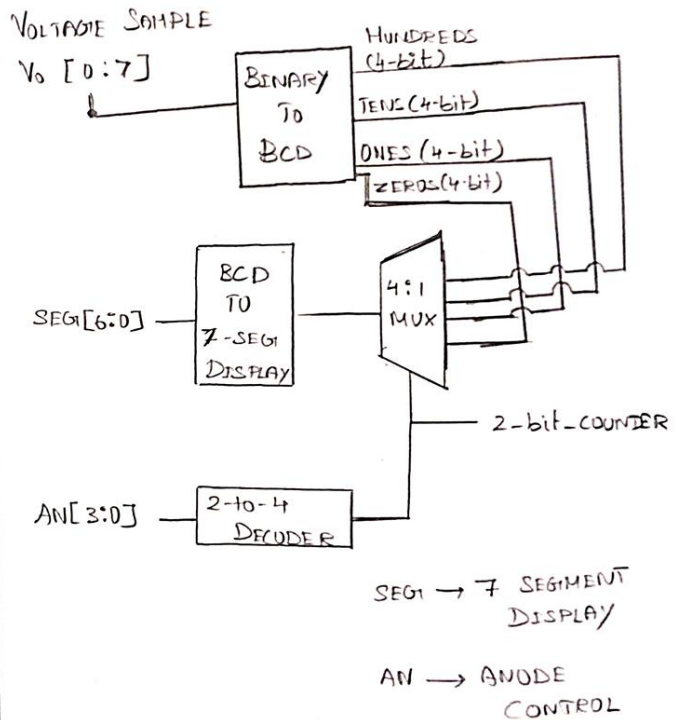
Encoded automatically by synthesis tool.



FAULT COUNTER (OVER AND UNDER VOLTAGE)



SEVEN SEGMENT DISPLAY



6. Simulation Results

Simulation was performed using Vivado XSIM with 100 MHz clock.

Test Cases

Switch Value	Decimal	State	LED Output
E6	230	NORMAL	Green
EB	235	NORMAL	Green
FA	250	OVER	Red
96	150	UNDER	Blue
E6	230	NORMAL	Green

Observations

- System correctly enters OVER state when voltage > 250.
- System correctly enters UNDER state when voltage < 210.
- System returns to NORMAL when voltage returns to nominal range.
- No oscillatory transitions observed.

Simulation confirms correct FSM behavior.

7. RTL Schematic Analysis

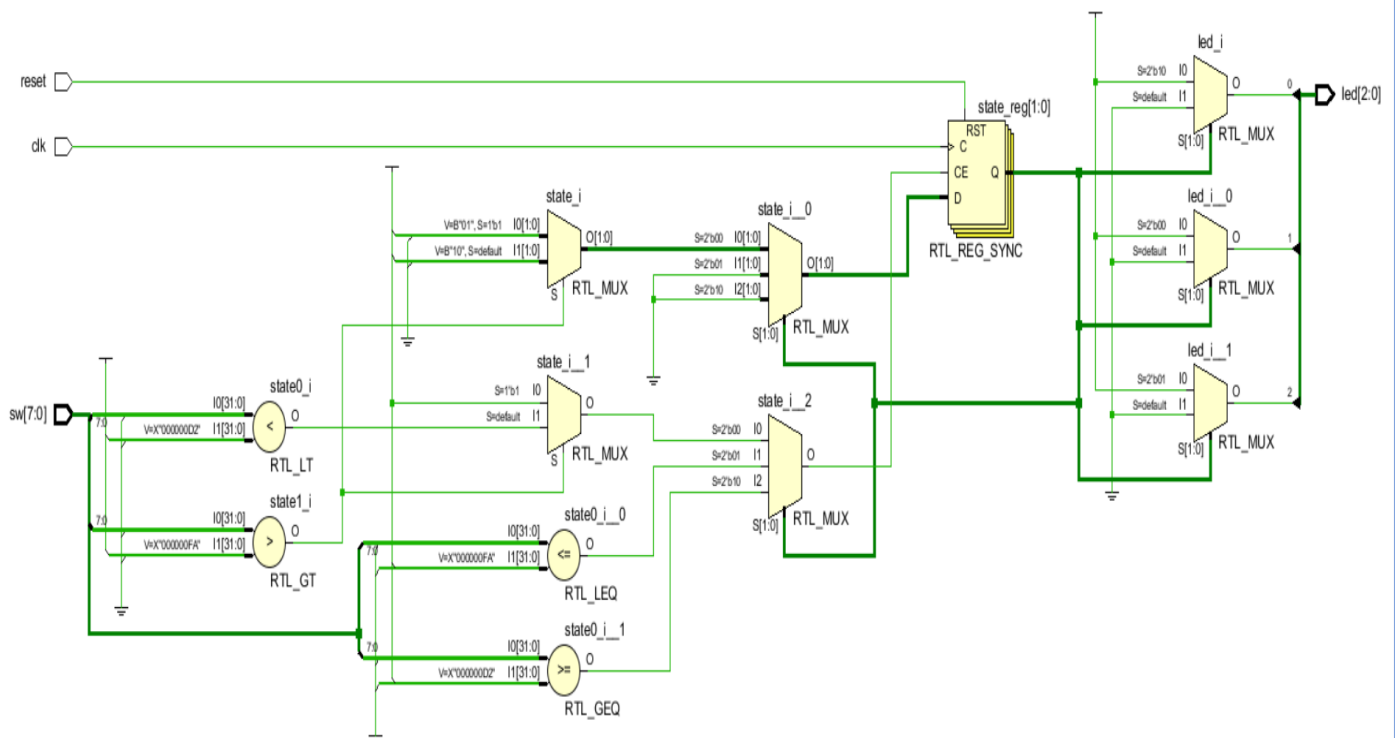
RTL schematic generated in Vivado shows:

- Comparator blocks ($<$, $>$, $\leq$, $\geq$)
- Multiplexer logic
- Synchronous state register (RTL_REG_SYNC)
- LED output mapping

Important Observations:

- No latches inferred
- Fully synchronous implementation
- Clean combinational logic
- Minimal logic depth

This confirms proper digital design methodology.

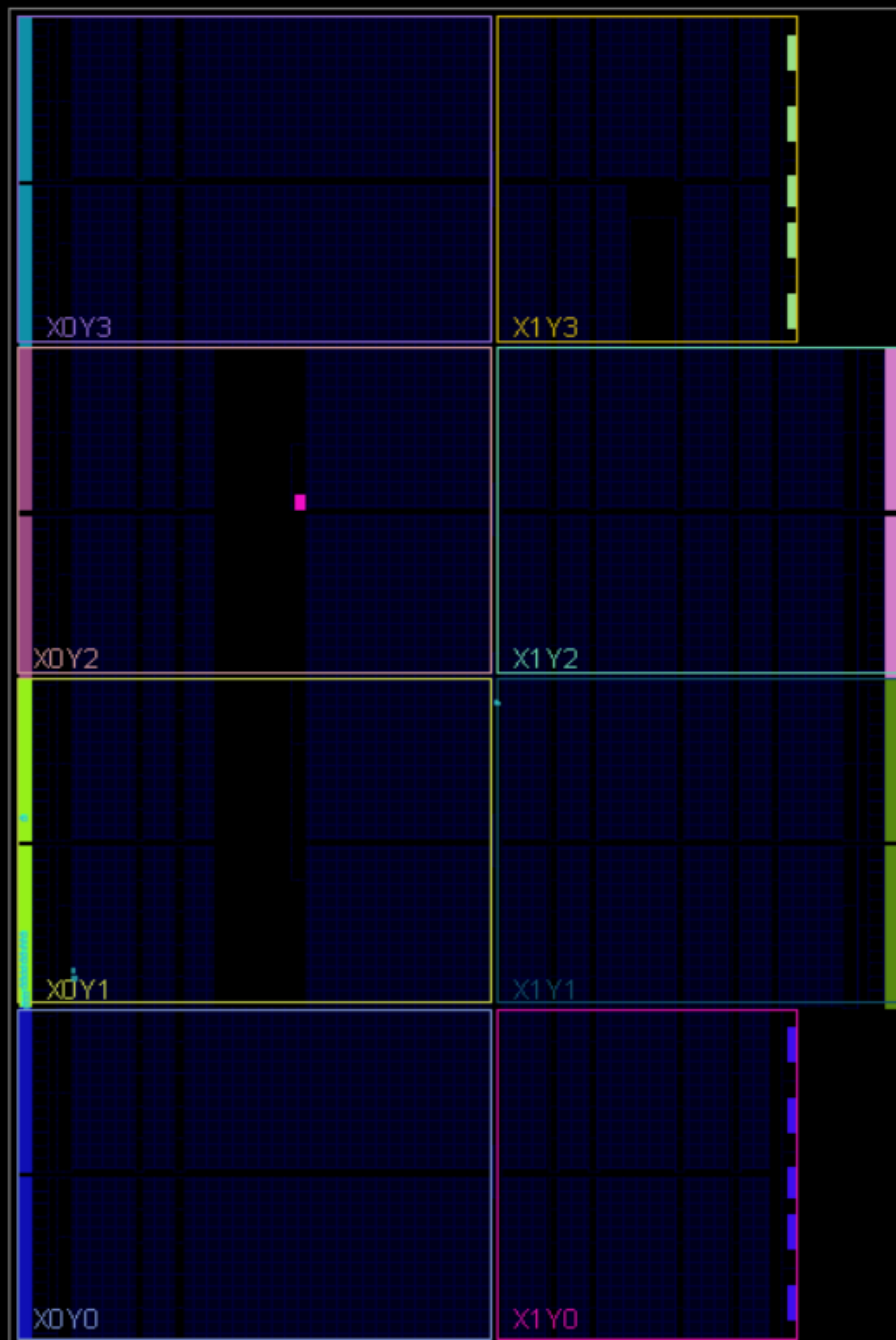


8. Device Utilization and Floorplanning

The floorplan indicates:

- Very small FPGA area utilized
- No routing congestion
- Centralized placement
- Efficient slice usage

This demonstrates that the design is lightweight and resource-efficient.



9. Power Analysis

Total On-Chip Power = 0.349 W

Power Breakdown

Dynamic Power = 0.257 W (74%)

Static Power = 0.092 W (26%)

Dynamic Components:

- I/O Power = 0.191 W (dominant)
- Signal Power = 0.042 W
- Logic Power = 0.025 W

Junction Temperature = 26.6°C

Thermal Margin = 58.4°C

Interpretation

- Logic consumes very low power.
- I/O switching dominates dynamic consumption.
- System operates well within safe thermal limits.

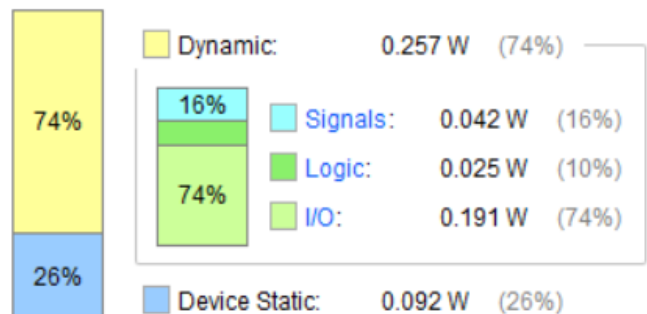
Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power:	0.349 W
Design Power Budget:	Not Specified
Power Budget Margin:	N/A
Junction Temperature:	26.6°C
Thermal Margin:	58.4°C (12.7 W)
Effective θ_{JA} :	4.6°C/W
Power supplied to off-chip devices:	0 W
Confidence level:	Low

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power



10. Hardware Implementation

The design was implemented on:

FPGA: Artix-7 XC7A100T (Nexys A7-100T)

Hardware Mapping:

- SW[7:0] → Voltage Input
- LED[2] → Over Voltage Indicator
- LED[1] → Normal Indicator
- LED[0] → Under Voltage Indicator

Bitstream was generated and programmed successfully. Hardware behavior matched simulation results.

14. Conclusion

The FPGA-based Adaptive Voltage Window Protection System was successfully designed and implemented using VHDL. The system accurately detects over-voltage and under-voltage conditions using comparator logic and a synchronous FSM. The Freeze-on-Fault mechanism enhances system stability by preventing threshold drift during abnormal conditions.

Simulation, RTL analysis, floorplanning, and power evaluation confirm correct and efficient implementation. The design demonstrates practical application of digital electronics concepts in FPGA-based protection systems.
